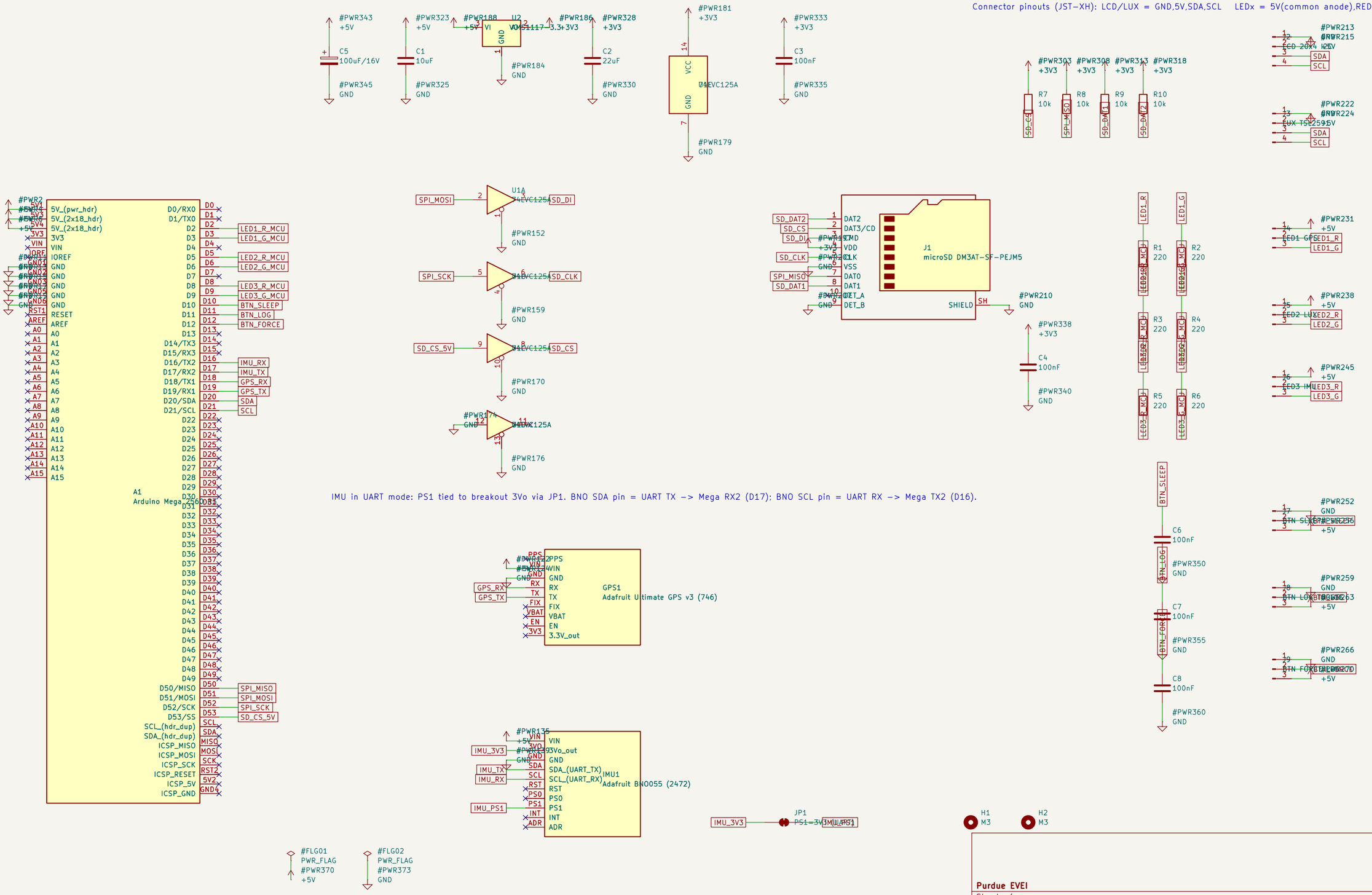
EVEI LIGHT METER — Arduino Mega 2560 carrier shield

microSD: on-board socket, 3.3V LDO, 74LVC125 shifts MOSI/SCK/CS 5V->3.3V; MISO (3.3V) goes straight to D50 (same as Adafruit 254 breakout).

Pin map from latest_version_of_lightmeter.ino: SD CS=D53 (HW SPI 50/51/52), GPS=Serial1 (D18/D19), IMU=Serial2 (D16/D17), I2C=D20/D21,
LEDs common-anode to 5V (active LOW): GPS D2/D3, LUX D5/D6, IMU D8/D9; buttons to GND with INPUT_PULLUP: sleep D10, log toggle D11, force log D12.

Connector pinouts (JST-XH): LCD/LUX = GND,5V,SDA,SCL LEDx = 5V(common anode),RED,GREEN BT

IMU in UART mode: PS1 tied to breakout 3Vo via JP1. BNO SDA pin = UART TX -> Mega RX2 (D17); BNO SCL pin = UART RX -> Mega TX2 (D16).



Purdue EVEI

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Title: EVEI Light Meter — Mega 2560 Shield

Size: A3	Date: 2026-09-24	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1